

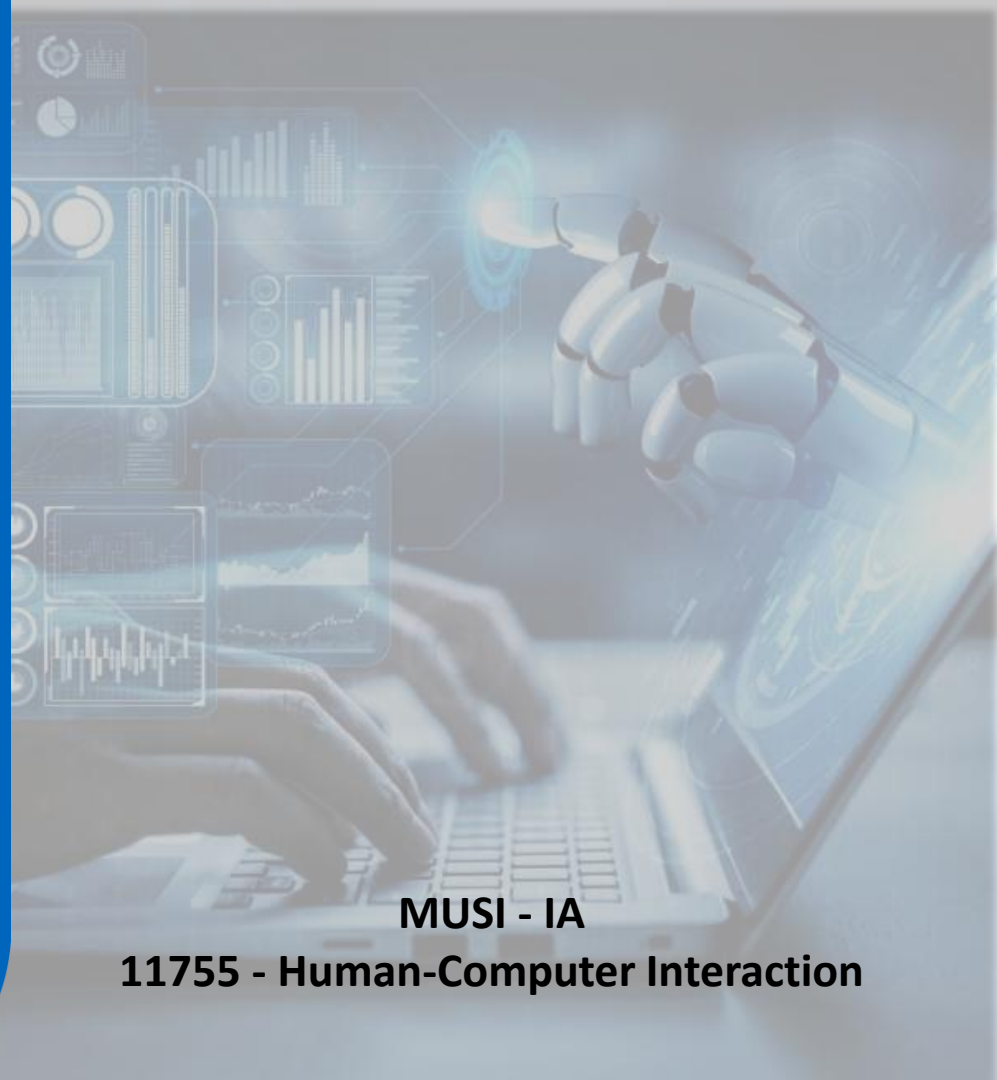


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03 - Interaction and Interface Design

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10-3-2025



MUSI - IA

11755 - Human-Computer Interaction

Interaction and User Interface design

- The disciplines of interaction design (IxD) and user interface (UI) design form the backbone of creating digital experiences that are intuitive, efficient, and inclusive.
- Interaction design focuses on shaping how users engage with systems through behaviors and workflows, while UI design emphasizes the visual and structural components that facilitate these interactions.
- Together, they bridge human needs with technological capabilities, ensuring products are both functional and emotionally resonant.



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1

Principles of Interaction Design (IxD)

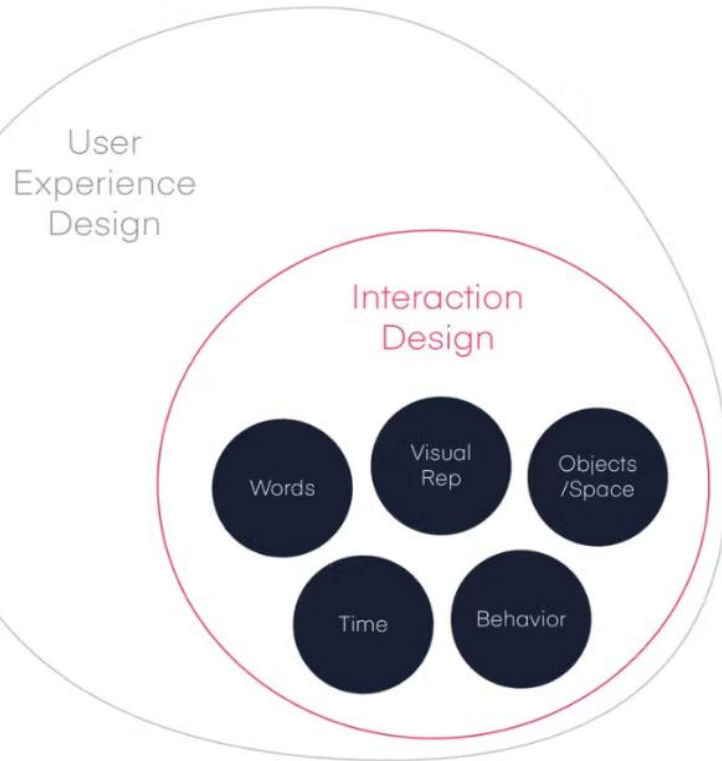
Interaction

- At a stripped-down level, **interaction** involves humans using their sensors and responders to monitor and control devices, machines, or systems that include computing technology.
- The internal states of the human and machine interact in a closed-loop system through controls and displays (the machine side) and motor-sensory behaviors (the human side). Inputs, or input devices, are inputs to the machine that are controlled or manipulated by human outputs.
- Most commonly, human outputs are limbs – the fingers, hands, arms, legs, feet – but speech, articulated sounds, eye motions, blinks, winks, head nods, breath, brain signals, and so on, can also serve as human outputs.

Interaction Design (IxD)

- **Interaction design** can be understood in simple (but not simplified) terms as the design of the interaction between users and products. The goal of IxD is to create products that enable users to achieve their objective(s) in the best way possible.
- IxD involves understanding how users interact with technology and designing interfaces that facilitate these interactions in intuitive and fulfilling ways.
- IxD encompasses various activities, including designing user flows, wireframes, and prototypes that facilitate seamless and meaningful user experiences.
- IxD operates at the intersection of human psychology, technology, and aesthetics, guided by principles that ensure interfaces align with user expectations and cognitive patterns.

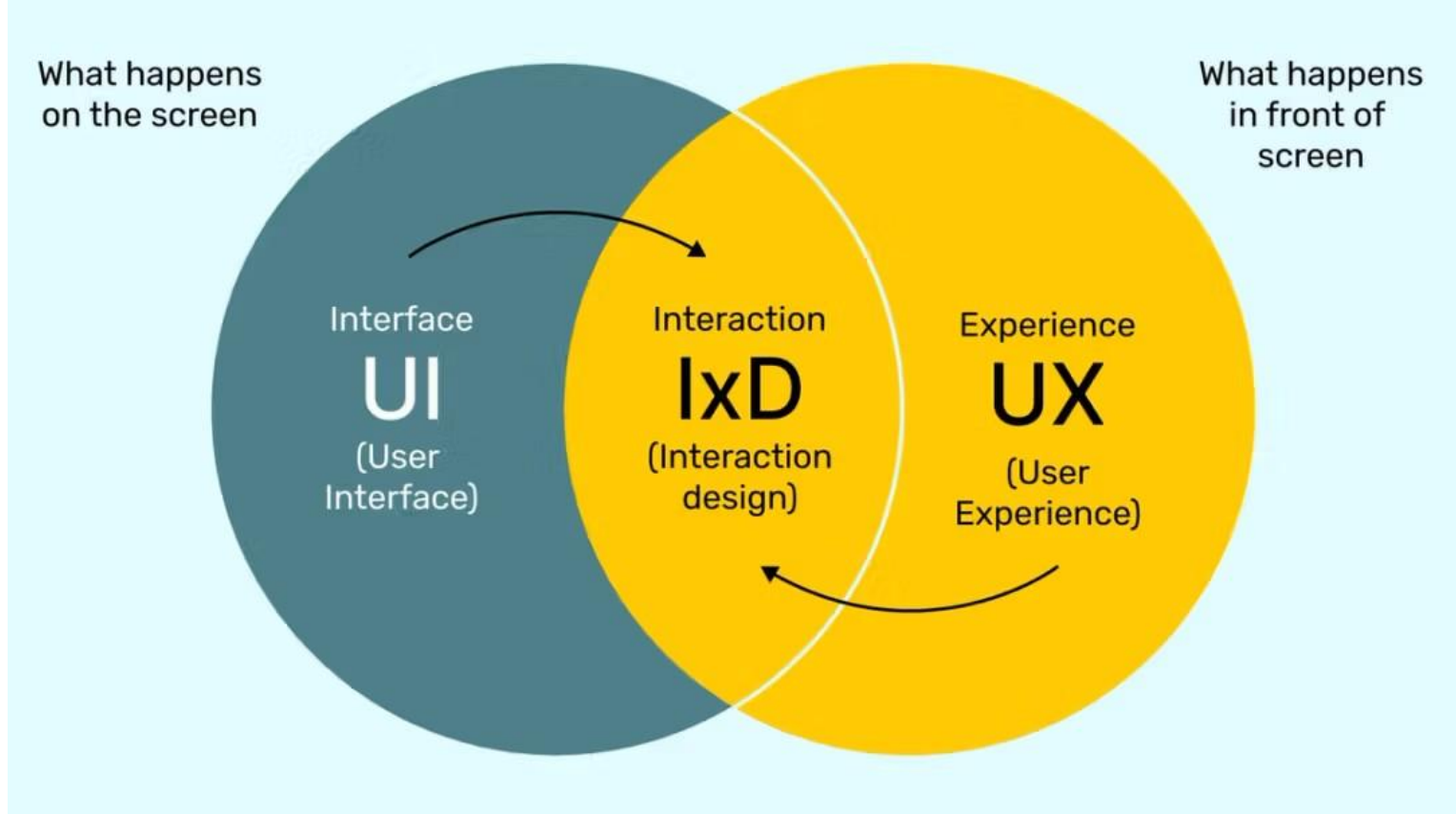
Interaction Design (IxD)



- While interaction design (IxD), user experience (UX) design, and user interface (UI) design are related fields, they each have distinct focuses.
- Interaction design primarily concerns creating interactive elements and ensuring these interactions facilitate effective user experiences.
- UI design focuses on the visual elements of a product's interface, such as layout, typography, and color schemes.

○ ["What Is an Interaction Designer?"](#)

Interaction Design (IxD)





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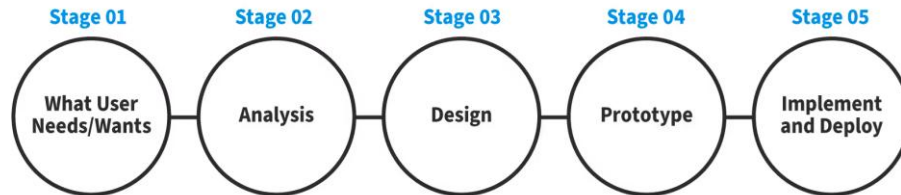
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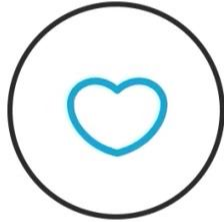
Interaction Design Process

Interaction Design Process

- The IxD process is the methodological approach that designers use to create solutions centered on users' needs, aims and behavior when they interact with products.
- The IxD process has **5 stages**: discovering user needs, analyzing them, designing a potential solution, prototyping it and then implementing and deploying it.

General Flow of Interaction Design Process





Empathise



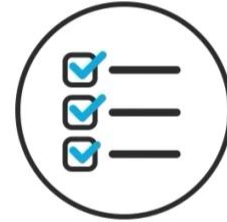
Define



Ideate



Prototype



Test

IxD Process – Stage 1 - Understand user needs and wants

- This initial phase involves researching and understanding users' needs, goals, and contexts of use.
- Designers gather insights through **user research**, which can include **interviews, surveys, observations, and personas**.
- Two useful research exercises include:
 - **Contextual inquiry:** This approach allows designers to immerse themselves in the user's environment. For instance, if designing a healthcare app, designers could observe how patients and doctors interact with existing systems.
 - **User interviews and surveys:** These exercises help designers gather direct insights from their target audience. For a retail app, a designer might explore shopping habits and pain points.

IxD Process – Stage 2 - Analyze user research findings

- This phase involves setting clear objectives, specifying user requirements, and establishing design principles that will guide the project.
- Creating user stories and scenarios and conducting task analyses can help clarify how proposed solutions fit into the users' lives.
- Here are two ways designers can analyze and organize their findings:
 - **Personas:** These are created based on research and should be as detailed as possible. For a project like a new educational platform, personas might include students, teachers, and administrators with distinct needs and goals.
 - **Journey mapping:** Developing journey maps helps designers visualize a user's interactions with a product. This can highlight friction points in current systems, like booking a service or completing a purchase.

IxD Process – Stage 3 - Design a potential solution

- In this phase, designers brainstorm, sketch and design potential solutions.
- This step often involves creating low-fidelity wireframes, flow diagrams and sketches that explore different design options and how users will interact with the system.
- Designers also think about the look and feel of the product, as well as the logic and structure of the interaction.
- Here are two approaches to designing at this point in the design process:
 - **Sketching:** Rapid, freehand drawing helps designers quickly visualize ideas, layout arrangements, and user interfaces.
 - **Storyboarding:** This involves creating a series of drawings to depict the user journey or specific interactions with a product.

IxD Process – Stage 4 - Create prototypes from designs

- Prototyping is an extension of the design phase, where ideas are transformed into tangible artifacts that can be tested and evaluated.
- Prototypes range from low-fidelity sketches to high-fidelity, interactive simulations that closely mimic the final product.
- Here are the two main types of prototypes:
 - **Low-fidelity prototyping:** Paper prototypes or basic digital mockups can demonstrate ideas and design quickly and easily. This approach is cost-effective and allows for rapid iteration based on initial feedback.
 - **High-fidelity prototyping:** For a deeper test of interactions, designers can create high-fidelity prototypes using tools like Sketch or Figma. For example, an e-commerce site design would benefit from detailed prototypes that simulate the shopping experience.

IxD Process – Stage 5 - Implement and deploy the final design

- Once the design has been refined through multiple iterations of testing and feedback, it moves into implementation, where developers build the interaction design into the product.
- Designers work closely with developers to ensure the design's fidelity and to adjust the interaction design as needed based on technical constraints or new insights.
- Carry out these two activities at this stage of the design process:
 - **Collaboration with developers:** Work closely with developers during the implementation phase to ensure the design translates well into the final product.
 - **Post-launch evaluation:** After launch, designers should collect and analyze user feedback. Tools like Google Analytics or user feedback platforms can provide insights into how well the product meets user needs.



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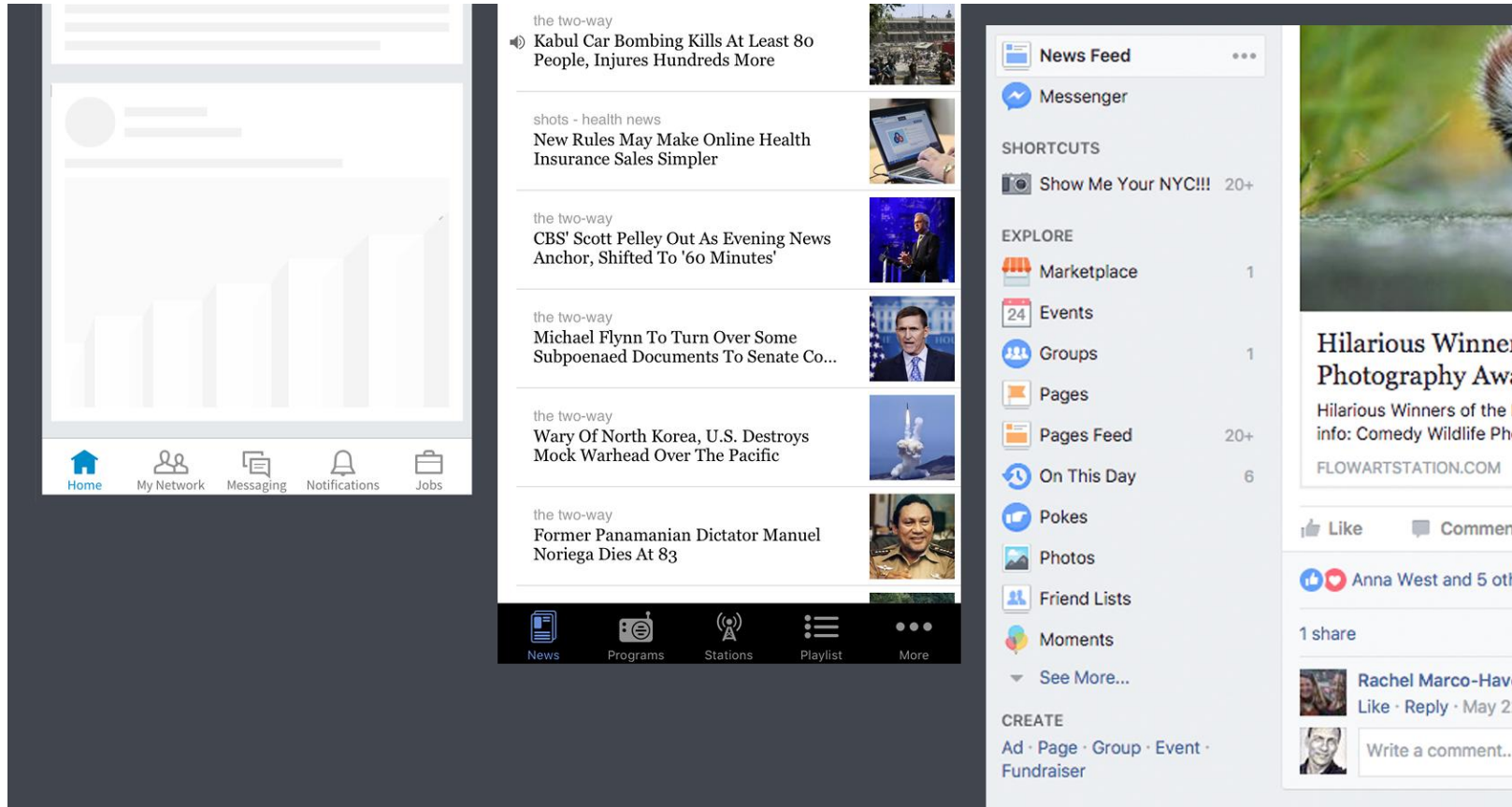
Interaction Design Principles

IxD Principles - Discoverability

- **Discoverability ensures users can immediately perceive available actions within an interface.** As Norman [2] notes, if a feature isn't discoverable, it effectively doesn't exist for the user.
- It must be made clear as a bell what actions are possible in a UI in less than a microsecond—for example, labeling icons as a best practice.
- Unlabeled icons are tantamount to throwing roadblocks into a user's path because they'll have to stop to decipher meaning. You're interrupting the “flow.”

[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Discoverability



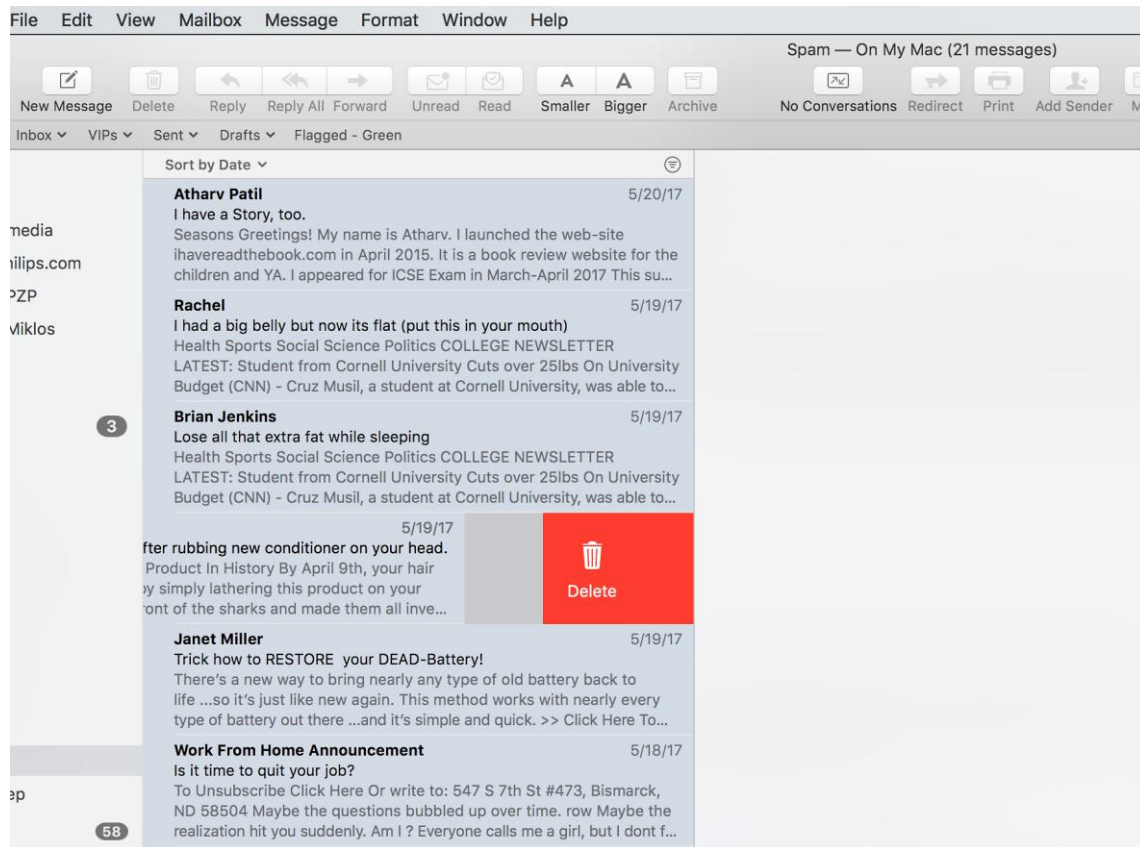
[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Signifiers

- **Signifiers—visual or auditory cues like labeled buttons, highlighted text, or animated hints**—act as perceptible signals that communicate affordances (the possibility of an action on an object).
- Very much tied to discoverability, effective use of signifiers ensures that affordances are clearly indicated in the UI.
- Effective signifiers eliminate ambiguity, enabling users to navigate interfaces without prior training.

[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Signifiers



[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Feedback

- After signifiers, the next most important communication to the user is feedback. **Feedback means: explicit information about the impact of the action.**
- It also means constant visibility of system status, i.e., the system should make sure that:
 - Users are kept informed about what's going on.
 - There is continuous information about the results of their actions and the current state of the product (system).
 - There is no confusion in the mind of a user as to the state of the product, i.e., processing, loading, searching, uploading, etc., or some other state change.
 - After an action has been executed, it's easy to determine the new state.

IxD Principles - Feedback



[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Mappings

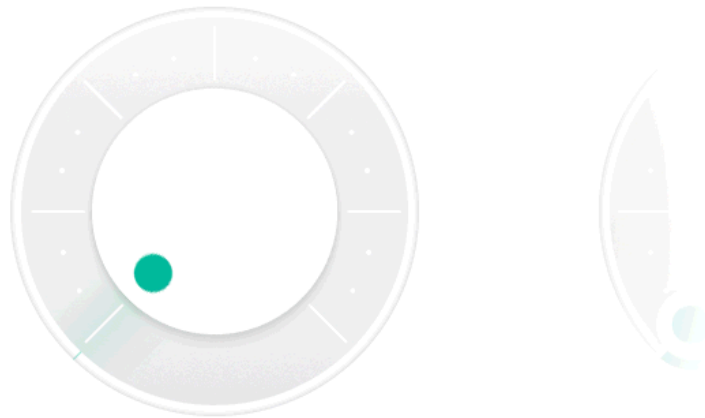
- **Mappings** define the relationship between controls and their effects in the world, leveraging spatial, cultural, or biological analogies to create intuitive connections.
- The principle of feedback is a close cousin of mapping because the two principles work together to create a seamless experience.
- Nearly all artifacts need some kind of mapping between controls and effects—in the case of interface design, it's the relationship between a control and its resulting function.
- Natural mapping—taking advantage of physical analogies and cultural standards—leads to an immediate understanding. For example, to increase some value with a slider component, swipe a slider to the right, to decrease it, swipe left.

IxD Principles - Mappings



[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Mappings



Level

IxD Principles - Conceptual and Mental Models

- **Great interaction design presents all the information a user needs to create a good conceptual model of the system**, thereby leading to their understanding and a sense of control.
- A conceptual model enhances both discoverability and evaluation of results when using the system.
- **Mental models—or cognitive maps—are the images in a user's mind that inform their expectation of a certain interaction** and how something works in the real world.
- Cognitive maps are internal representations of our physical environment, particularly associated with spatial relationships. By effectively using the user's mental model, interaction designers can create interaction design systems that “feel” intuitive.

IxD Principles - Conceptual and Mental Models

- The designer invents a conceptual model—the design model, and this is how the designer intends the device or software to work.
- The only way to communicate this model to users is to implement it through a UI.
- Users then interact with the conceptual model and create their own mental model of how something works.
- Designers can't tell users how they intended an app to work—they need to design a UI appropriately in order to clearly communicate its purpose. In other words, make the interface easy to understand and use.

[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Conceptual and Mental Models

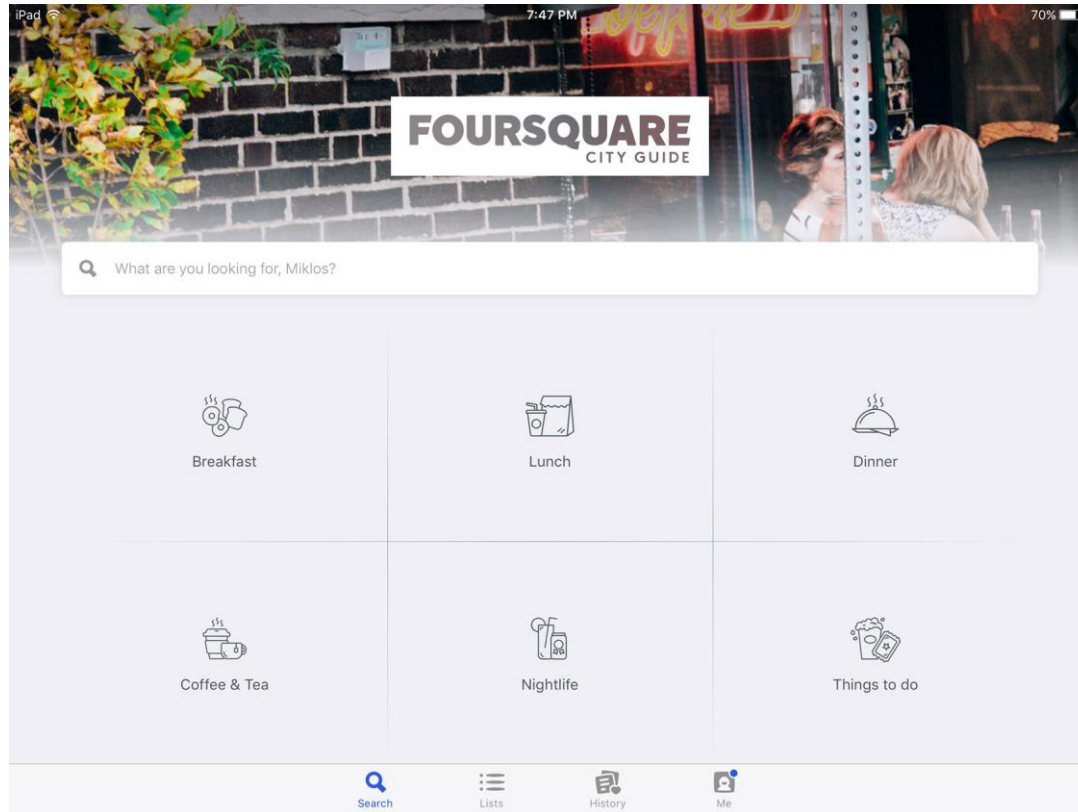


[Boost Your UX With These Successful Interaction Design Principles](#)

IxD Principles - Constraints

- **The design concept of constraints is a way to determine how to restrict the kind of user interaction that can take place in a given moment.**
- Interaction design constraints help efficient interaction by providing “guide rails” for users—almost like a guiding hand steering the interactions that can occur.
- Constraints in design make sure only specific things are enabled, or even visible, in order to guide the user towards certain interactions.
- Constraints put in place correctly also help reduce the chance of user error. Great interaction design doesn’t put every available option on the interface, but restricts it to what’s most important in any given moment.

IxD Principles - Constraints



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1.3

Interaction Design Dimensions

IxD's five dimensions

IxD's five dimensions were first defined by a professor at London's Royal College of Art, Gillian Crampton Smith, and a senior interaction designer, Kevin Silver.

Interaction Design Foundation - IxDF. [“What is Interaction Design \(IxD\)?”](#)

5 DIMENSIONS OF INTERACTION DESIGN

[illegible]

INTERACTION DESIGN
FOUNDATION

INTERACTION-DESIGN.ORG

IxD's five dimensions

The dimensions represent the aspects an interaction designer considers when designing interactions:

- Words (1D) encompass text, such as button labels, which help give users the right amount of information.
- Visual representations (2D) are graphical elements such as images, typography and icons that aid in user interaction.
- Physical objects/space (3D) refers to the medium through which users interact with the product or service—for instance, a laptop via a mouse, or a mobile phone via fingers.

IxD's five dimensions

The dimensions represent the aspects an interaction designer considers when designing interactions (continuation):

- Time (4D) relates to media that changes with time, such as animations, videos and sounds.
- Behavior (5D) is concerned with how the previous four dimensions define the interactions a product affords—for instance, how users can perform actions on a website, or how users can operate a car.
- Behavior (5D) also refers to how the product reacts to the users' inputs and provides feedback.



5 DIMENSIONS OF INTERACTION DESIGN



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1.4

Interaction Design Patterns

IxD Patterns

- An IxD pattern is a general repeatable solution to a commonly-occurring usability problem in interface design or interaction design.
- An ID pattern usually consists of the following elements:
 1. **Problem:** Problems are related to the usage of the system and are relevant to the user or any other stakeholder that is interested in usability.
 2. **Use when:** a situation (in terms of the tasks, the users and the context of use) giving rise to a usability problem.
 3. **Principle:** a pattern is usually based on one or more ergonomic principles such as user guidance, or consistency, or error management.
 4. **Solution:** a proven solution to the problem. A solution describes only the core of the problem, and the designer has the freedom to implement it in many ways.

IxD Patterns (Continuation)

- An ID pattern usually consists of the following elements (continuation):
 5. **Why:** How and why the pattern actually works, including an analysis of how it may affect certain attributes of usability. The why should describe which usability aspects should have been improved or which other aspects might suffer.
 6. **Examples:** Each example shows how the pattern has been successfully applied in a real-life system. This is often accompanied by a screenshot and a short description.
 7. **Implementation:** Some patterns provide implementation details.

IxD Patterns - Wide-used interaction patterns

- **Navigation patterns:** Help users understand the product, easily find needed functionality (samples: menu, tabs, breadcrumbs).
- **Getting input patterns:** Help users interact with the system by filling the data and receiving results (for instance: text fields, select boxes, drag and drop, pickers, etc).
- **Dealing with data patterns:** Organize content for quick review, easy search and processing (samples: galleries, tables, cards).
- **Social patterns:** Optimise users' communication between each other online (chats, boards, testimonials, follow).

IxD Patterns - How to use patterns

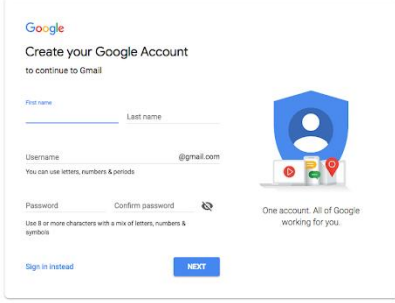
You are creating an application that requires users to enter their data in order to perform a mathematical calculation. The data should be provided in numerical values only.

1. **What is the problem?** Users are entering incorrect data, which causes mistakes and system inability to process the information.
2. **When does the problem appear?** Users can enter the data in a wrong format, which will cause the system mistake. As a result — users won't be able to make a correct calculation.
3. **Which principle to apply?** Preventing User Errors. You should minimize the number of conditions wherever there is a possibility for the user to make a mistake.
4. **How to solve the problem (Solution)?** Restrict user from entering wrong data and implement error message.

IxD Patterns - How to use patterns

- 5. Why we need to solve it?** By removing a system mistakes we will increase system credence among the users.
- 6. Was the pattern used before for a similar problem?** Google use hints to prevent users from making a mistakes. You can also use numbers-only keyboard or an input mask.
- 7. How to implement it?** By locking invalid keyboard characters we keep the user from entering wrong data (an input Mask).

Kateryna, Qbstudio, [“Interaction Design Patterns”](#)



The screenshot shows the Google Account creation interface. It features the Google logo at the top, followed by the heading 'Create your Google Account' and the subtext 'to continue to Gmail'. The form includes input fields for 'First name', 'Last name', 'Username' (with a placeholder '@gmail.com'), and 'Password'. A note below the password field states 'You can use letters, numbers & periods'. A 'Confirm password' field is also present, accompanied by an eye icon for toggling visibility. A 'Sign in instead' link is located at the bottom left of the form. On the right side, there is a blue shield icon with a white person silhouette and a red location pin, with the text 'One account. All of Google working for you.' below it. At the bottom right, there is a blue 'NEXT' button. The footer contains 'English (United States)' on the left and 'Help Privacy Terms' on the right.



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2

Visual Design Elements and Principles

Visual Design

- Visual design is about creating and making the general aesthetics of a product consistent.
- To create the aesthetic style of a website or app, we work with fundamental elements of visual design, arranging them according to principles of design.
- These elements and principles together form the building blocks of visual design, and a firm understanding of them is crucial in creating a visual design of any product.
- You use visual design to create and organize elements to:
 - lead the user's eye to an item's functionality,
 - make the aesthetics consistent.



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2.1

Visual Design Elements

Visual Design Elements

- Any product — from software products such as websites and apps to hardware products such as toasters and hairdryers — can be broken down into fundamental elements of visual design.
- These elements are the basic tools that the visual designers use in their daily work, and having a basic understanding of them is definitely a prerequisite for the job.
- Users' first impressions typically form in 50 milliseconds. If they don't like what you show them in those critical moments, they will leave.

Visual Design Elements



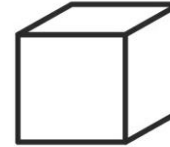
1. Lines



2. Shapes



3. Negative Space
/ Whitespace



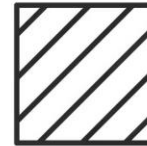
4. Volume



5. Value



6. Color

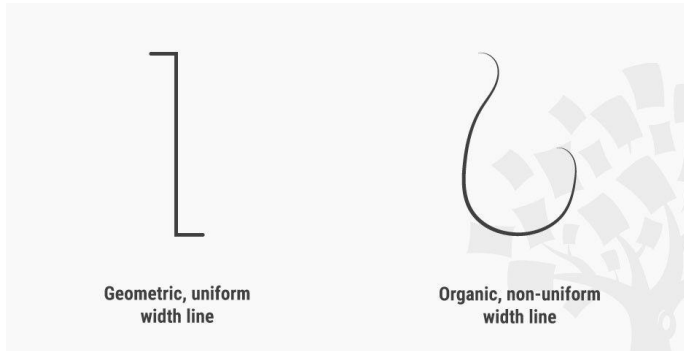


7. Texture

Visual Design Elements - Line

- Lines are strokes connecting two points, and the most basic element of visual design.
- We can use them to create shapes, and when we repeat them, we can form patterns that create textures.
- Although simple, lines can possess a large variety of properties that allow us to convey a range of expressions.

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



A line can also be implied: that is, suggested by forming an invisible connection between other elements.



Visual Design Elements - Negative/White Space

- Negative space (also known as white space) is the empty area around a (positive) shape.
- The relation between the shape and the space is called figure/ground, where the shape is the figure and the area around the shape is the ground.
- Negative space is just as important as the positive shape itself — because it helps to define the boundaries of the positive space and brings balance to a composition.

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



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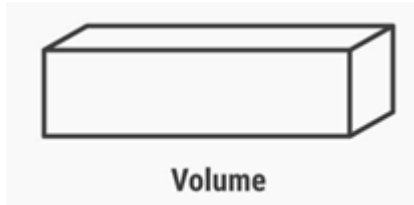
WWF's logo doesn't explicitly draw out the entire panda: it cleverly uses negative (white) space around the black shapes to suggest the rest of the panda



Visual Design Elements - Volume

- Volume applies to visuals that are three-dimensional and have length, width and depth.
- We rarely use volume in visual design, because most digital products end up being viewed on a 2D screen, although some apps and websites do use 3D models and graphics.

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)

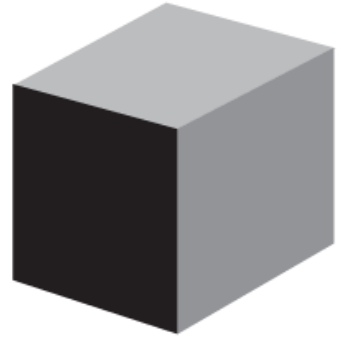
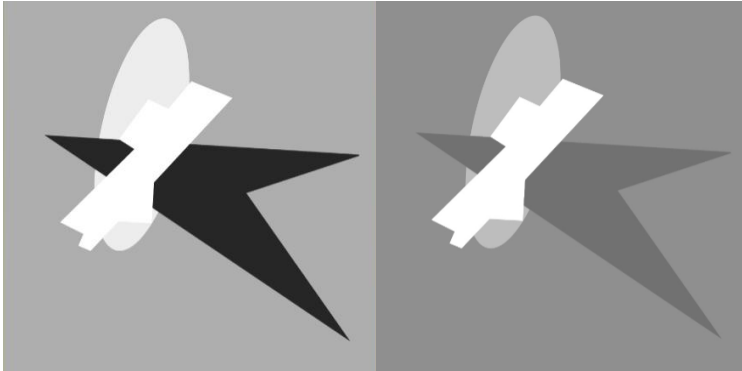


Volume has 3 dimensions: length, width and depth. This image is a simulation of volume in 2D graphics

Visual Design Elements - Value

- Value, quite simply, describes light and dark.
- A design with a high contrast of values creates a sense of clarity, while a design with similar values creates a sense of subtlety.
- We can also use value to simulate volume in 2D, for instance, by using lighter values where the light hits the object and darker values for shadows.

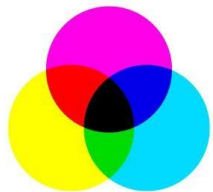
Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



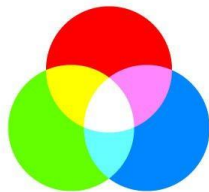
Visual Design Elements - Color

- Color is an element of light. Color theory is a branch of design focused on the mixing and usage of different colors in design and art.
- In color theory, an important distinction exists between colors that mix subtractively and colors that mix additively.
- The subtractive mix of colors in paint and print produces the CMYK color system. The additive mix of colors on digital screens produces the RGB color system.

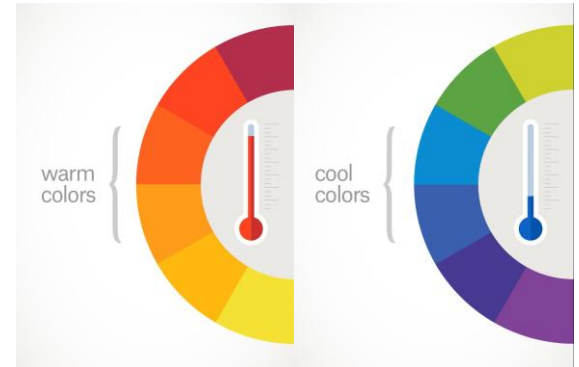
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Subtractive mix
(in print)



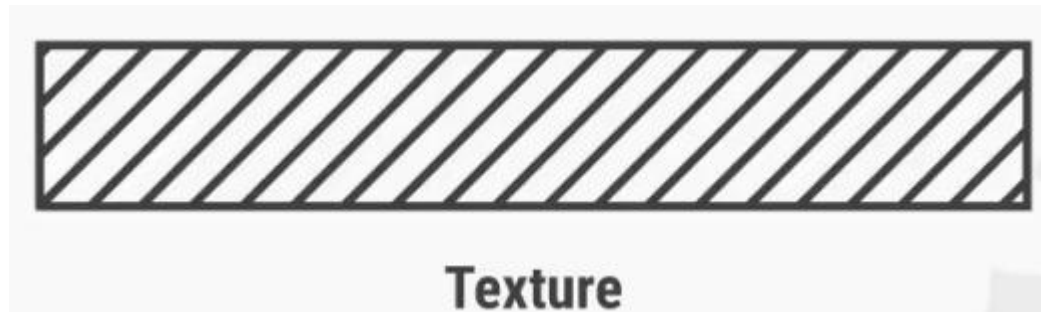
Additive mix
(on screen)



Visual Design Elements - Texture

- Texture is the surface quality of an object.
- You can work with two types of textures: tactile textures, where you can feel the texture, and implied textures, where you can only see — i.e., not feel — the texture.
- Visual designers will work with implied textures, since screens are unable to produce tactile textures.

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



The diagonal lines add a 'grip' effect to an otherwise 'smooth' rectangle.



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2.2

Visual Design Principles

Visual Design Principles

- The elements of visual design — line, shape, negative/white space, volume, value, color and texture — describe the building blocks of a product's aesthetics.
- On the other hand, the principles of design tell us how these elements can and should go together for the best results.
- Many of the following principles are closely related and complement one another.

Interaction Design Foundation - IxDF. ["The Key Elements & Principles of Visual Design"](#)

Visual Design Principles

Visual Design Principles



1. Unity



2. Gestalt



3. Hierarchy



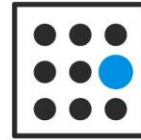
4. Balance



5. Contrast



6. Scale



7. Dominance

Visual Design Principles - Unity

- **Unity has to do with creating a sense of harmony between all elements in a page.**
- A page with elements that are visually or conceptually arranged together will likely create a sense of unity..
- When we're designing websites, we can **make use of a grid for achieving a sense of unity, since elements organized in a grid will follow an orderly arrangement.**

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



Visual Design Principles - Gestalt

- **Gestalt refers to our tendency to perceive the sum of all parts as opposed to the individual elements.**
- We tend to perceive the overall shape of an object first, before perceiving the details (lines, textures, etc.) of the object.
- **Gestalt is important, for instance, in making separate sections of a website distinct by increasing the white space between them.**

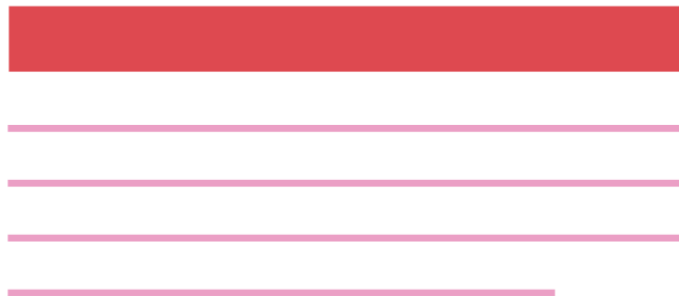
Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



Visual Design Principles - Hierarchy

- **Hierarchy shows the difference in importance of the elements in a design.**
- **Color and size are the most common ways we can create hierarchy** — for instance, by highlighting a primary button, or using larger fonts for headings.
- Items that appear at the top of a page or app also tend to be viewed as having a higher hierarchy than those appearing below.

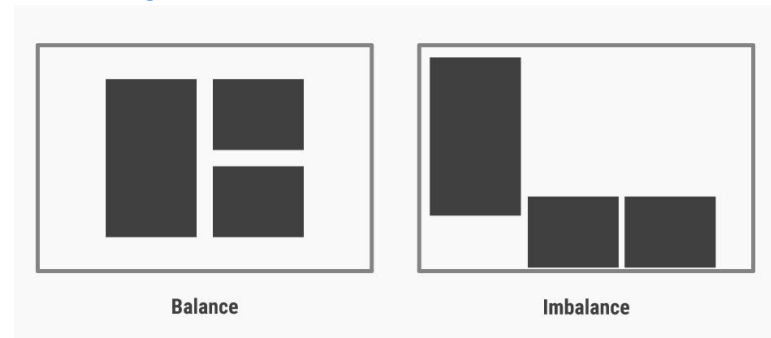
Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



Visual Design Principles - Balance

- **Balance is the principle governing how we distribute the elements of a design evenly.** Balanced designs tend to appear calm, stable and natural, while imbalanced designs make us feel uneasy.
- **Balance can be achieved by having symmetry in the design** (for instance, having a webpage with centralized text and images).
- We can also achieve **balance without symmetry when we arrange differently sized elements in a way that results in unity.**

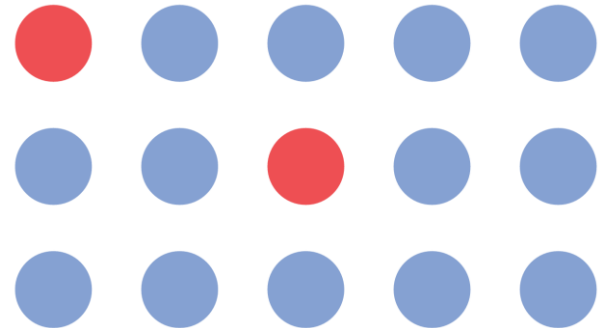
Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



Visual Design Principles - Contrast

- **We use contrast to make an element stand out by manipulating differences in color, value, size and other factors.**
- Designers often use the color red to make certain elements stand out. On the other hand, green is often something used (at least in Western design) in positive actions such as “Go” and “Accept”.
- **Use a color-contrast checker to ensure that your content can be read by all your target users.**

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)



Visual Design Principles - Scale

- **Scale refers to the relative sizes of certain elements in a design, in comparison to other elements.**
- Scale can also establish a visual hierarchy, making the more prominent elements seem more important than the others.
- **Use no more than 3 different sizes.**
- **Emphasize the most important aspect by making it the biggest.**

Interaction Design Foundation - IxDF. ["The Key Elements & Principles of Visual Design"](#)



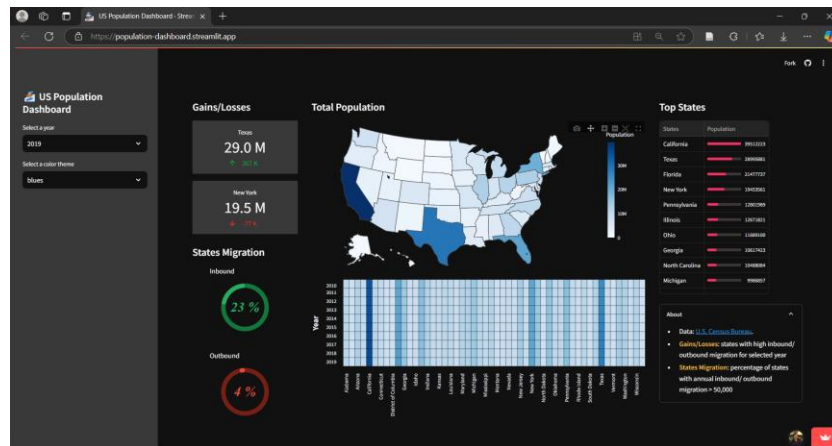
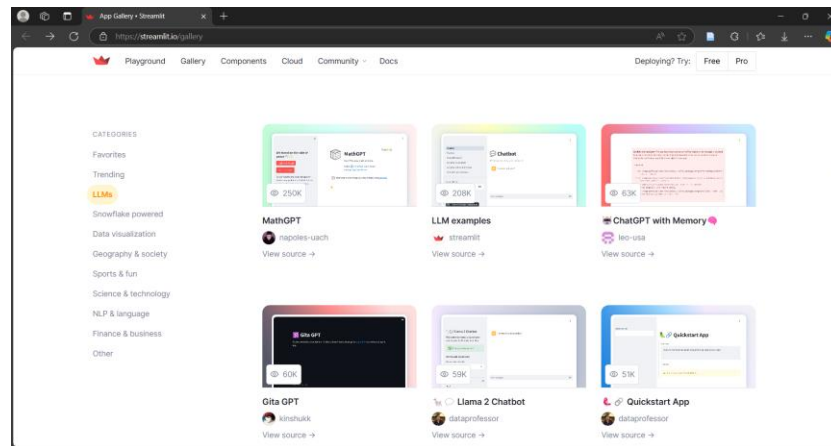
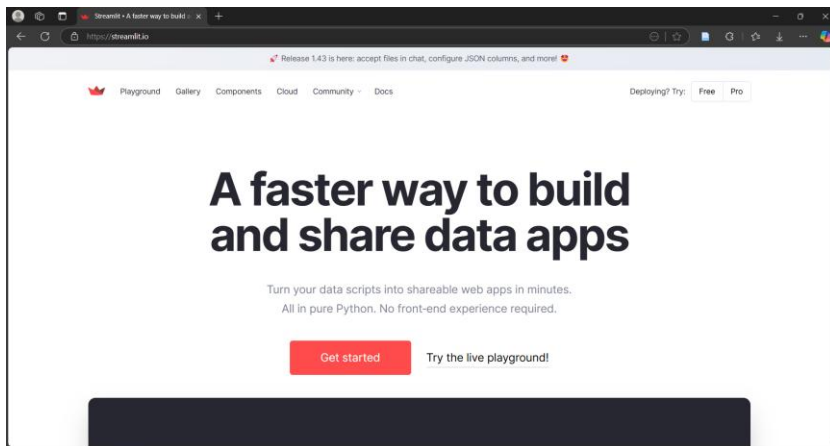
Visual Design Principles - Dominance

- **Dominance creates focus on a single element.**
- We can use color, shape, contrast, scale, and/or positioning to achieve this.
- For instance, most websites have a main “hero” image, which uses dominance to appeal to users, drawing them to it naturally.
- When working in visual design, **we should ensure that we use dominance while still maintaining the unity and balance of websites**

Interaction Design Foundation - IxDF. [“The Key Elements & Principles of Visual Design”](#)

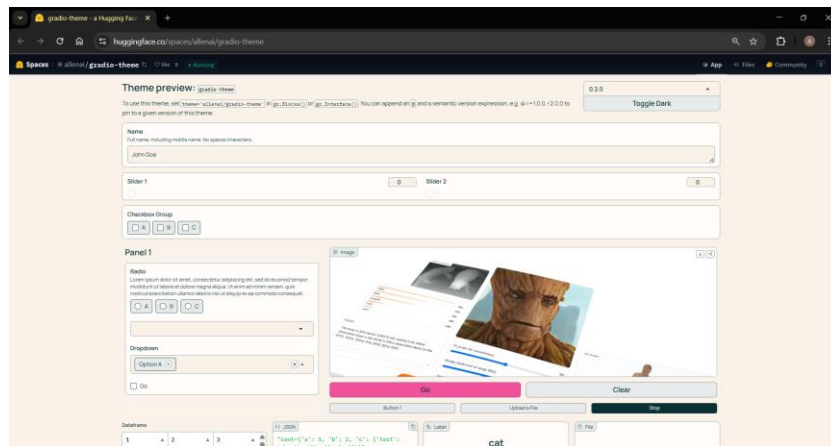
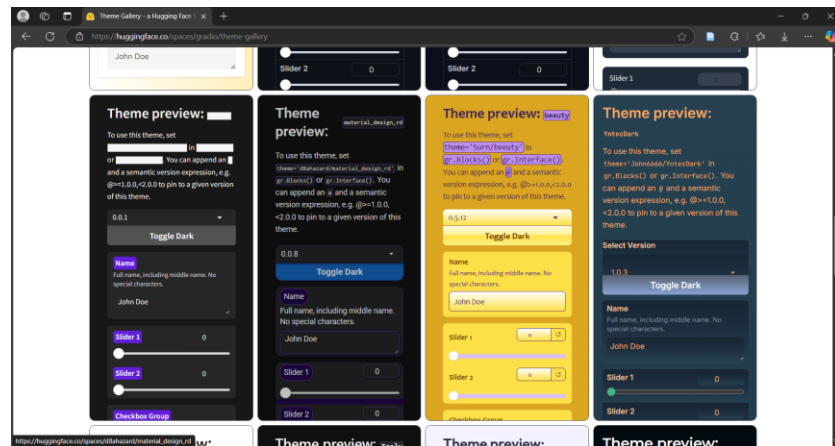
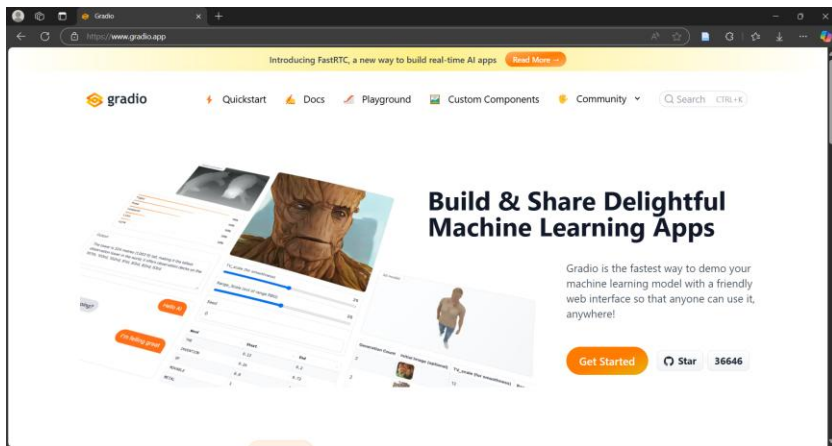


Visual Design - Python



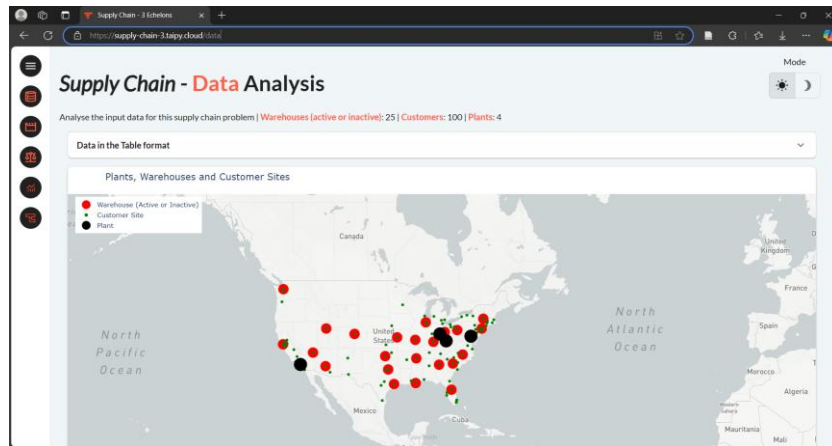
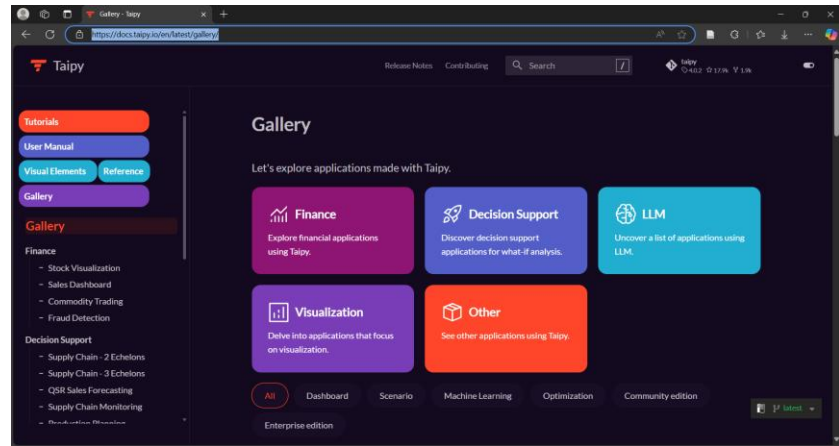
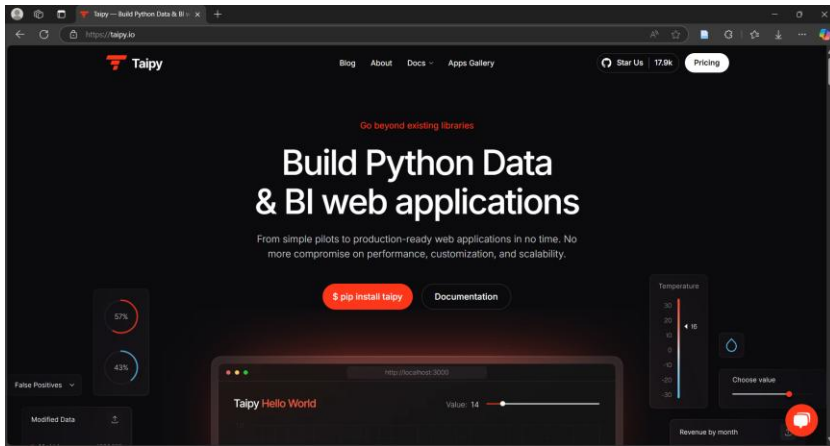
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Visual Design - Python



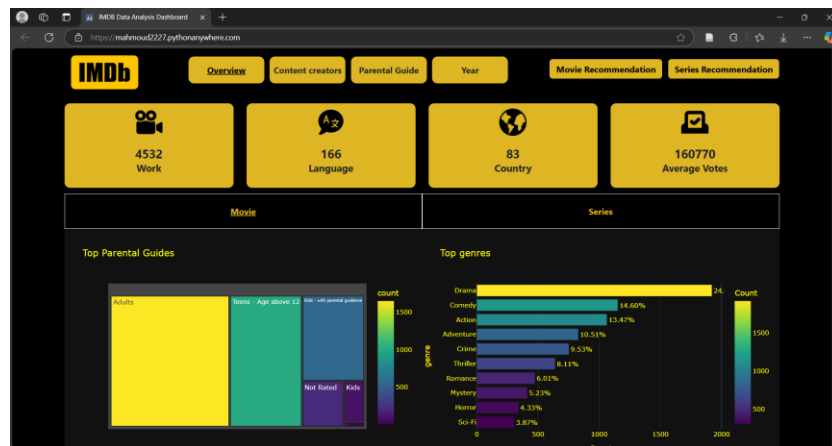
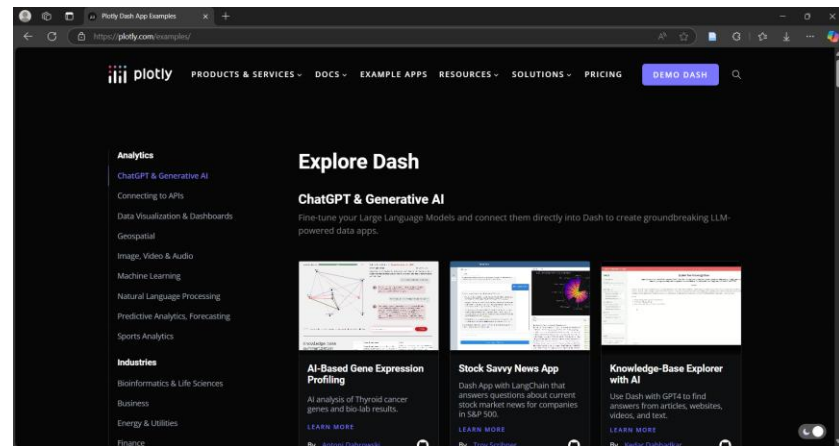
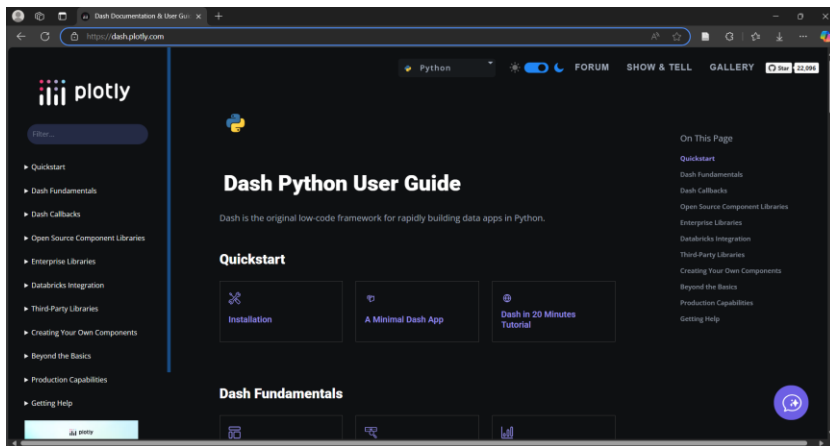
Gradio

Visual Design - Python



Taipy

Visual Design - Python



Dash

Recommended Readings:

- [1] A. Dix, J. Finlay, G. D. Abowd, and R. Beale, Human-computer interaction, 3rd edition. Pearson Education Limited, 2004. (Chapter 7)
- [2] D. A. Norman, The design of everyday things, Rev. and Expanded edition. Basic Books, 2013. (Chapters 2 and 3)



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